MIT 8.012 Problem Set 4 - Problem 3.14 (K&K)

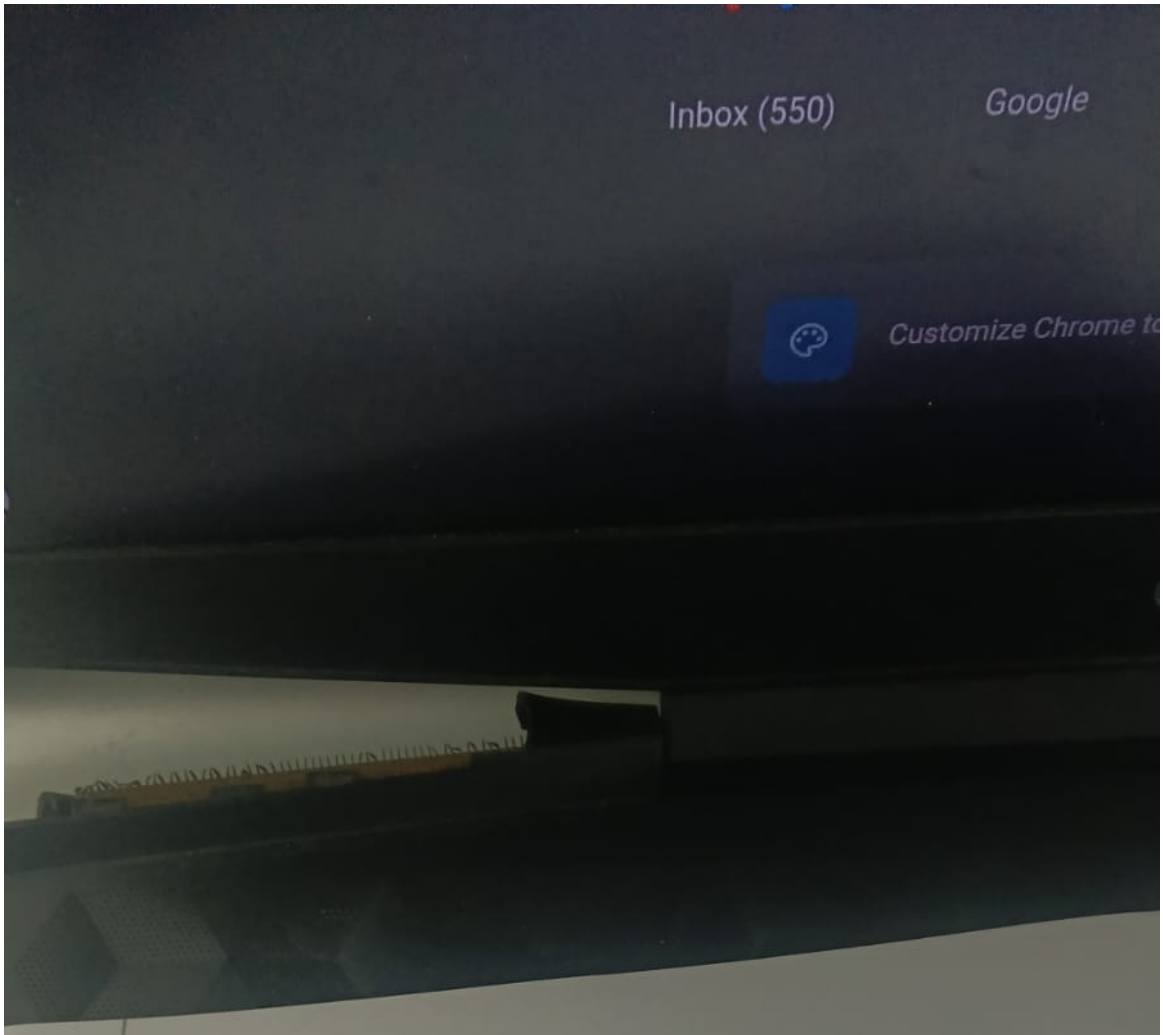
3.14 N men, each with mass m, stand on a railway flatcar of mass M. They jump off one end of the flatcar with velocity u relative to the car. The car rolls in the opposite direction without friction.

a. What is the final velocity of the flatcar if all the men jump at the same time?

b. What is the final velocity of the flatcar if they jump off one at a time? (The answer can be left in the form of a sum of terms.)

c. Does case *a* or case *b* yield the largest final velocity of the flat car? Can you give a simple physical explanation for your answer?

Solution (handwritten) follows on the next pages.



Problem 4.

Case a) All men jump at the same time

$$(M+mN)\vec{v_i} = (mN)(\vec{v_F} + \vec{u}) + M\vec{v_F}$$

$$(M+mN)\vec{v_i} = \vec{u}mN + \vec{v_F}(M+mN)$$

$$\vec{v_F} = \vec{v_i} - \frac{\vec{u}mN}{(M+mN)}$$

Case 1) They jump one at a time

$$(M + m(N-(n-1)))\vec{v_{n-1}}$$

$$\{ m(\vec{u} + \vec{v_n}) + (M + m(N-n))\vec{v_n}$$

$$\vec{v_n} = \frac{-m(\vec{u})}{(M+m(N-(n-1)))} + \vec{v_{n-1}}$$

The change in velocity is much higher in the second case, because momentum is conserved at every step and Δm is comparatively smaller than case 1, so Δv must be higher